Shanmugam Subramanian

Mechanical Engineer II | Rugged Radar & RF Hardware | Hyderabad, India

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SUMMARY

Mechanical engineer with 4 years of NPD and NPI experience designing rugged radar and RF hardware. Qualified products to IEC 60068 vibration and drop standards, designed IP68 (IEC 60529) radar chassis and radomes, and ran structural, thermal and CFD analysis. Experienced in configuration change control and contract-manufacturer setup.

EXPERIENCE

JMA Wireless — Hyderabad, India
Mechanical Engineer II May 2023 – Present

- Qualified antenna products to **IEC 60068 environmental standards**: harmonic & random vibration (-2-6/-64) with 80% first-pass correlation to physical tests, plus drop tests (-2-31).
- Executed **static structural analysis** with FOS targets of 1.5–3 (metals) & 2–5 (plastics).
- Owned end-to-end **NPD & NPI** for eight 5G Macro & Distributed Antenna Systems products for the US & Canada markets.
- R&D of radome profiles using **Ansys Fluent CFD** (2D/3D); Design of experiments on 135+ profiles achieving **12–18% wind load reduction**. Documented simulation workflows for team adoption & working toward patent filing for further optimization.
- Built Excel-based **hand-calculation tools** to estimate drag force from geometry, crimping force for pneumatic cylinder sizing, and snap-joint behavior in plastic part design.
- Designed pole, wall, ceiling, & tri-sector mounting brackets; gravity-assisted design **halved installation manpower**.
- Released RF PCBs, sheet metal, injection-molded, die-cast, CNC, EDM, & FDM parts with a **90% FAI pass rate**.
- Controlled design changes (**ECRs, FAIs, & SVPRs**) through Salesforce Propel **PLM** & SolidWorks **PDM** with the global team.
- Developed a **pneumatic-actuated pivot punching fixture** that automated a 2-person task to one operator and **improved yield from 60% to 95%** (saving $0.4 per failed cycle).
- Introduced topology-optimized FDM 3D-printed prototypes, **cutting 60 g per unit** while meeting stiffness requirements.
- Supported **contract manufacturing** setup in Chennai through BOM costing, tooling transfer, & operator training.
- Automation & process-improvement by developing RF PCB sub-assembly fixtures, rivet press-pull tools, & alignment jigs, achieving **40% rework reduction, 75% faster riveting, and cutting up to 20–50 mins per antenna** in the production line.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Design Engineer (R&D) Jul 2022 – Apr 2023

- Developed two automotive radar antenna chassis and radomes with **IP68 (IEC 60529)** ratings for ADAS applications.
- Enhanced chassis thermal properties by refining fin geometry, reducing **internal temperature by 6°C**.
- Reverse-engineered & locally manufactured an RF waveguide connector, reducing cost by 8× compared to imports.
- Lowered radome **cost by 8%** through material optimization without degrading dielectric performance.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Project Intern Mar 2022 – Jun 2022

- Updated automotive radar housing designs based on lead engineer feedback & prepared GD&T-compliant drawings.
- Implemented **snap-fit** radome design, reducing the radar BOM by two components and saving $0.3 per unit.

EDUCATION

Bachelor of Engineering — Mechanical Engineering GPA: 8.21 / 10
Alagappa Chettiar College of Engineering and Technology (Anna University) 2018 – 2022

PROJECTS AND COMPETITIONS

SAE Bicycle Design Competition — 2020 - 2021 SAE INDIA
Developed a planetary gear-based bicycle drivetrain achieving 15% higher torque than a typical bicycle.

Mini Trencher Design — APRIL/ MAY - 2021 3rd Year Project B.E.
Designed a low-cost mini agricultural trencher; reduced 5× cost & size through drivetrain & structure optimization.

Prodigy Design Competition— May 2021 NIT Trichy
Designed a regenerative adhesive rolling-belt mechanism for water-body waste removal within a 24-hour time frame; delivered CAD models & documentation, securing runner-up position among 50 teams.

SKILLS & SOFTWARE

Design: SolidWorks, AutoCAD, Fusion 360, Inventor, Onshape, GD&T, Tolerance stackup, 3D Printing
Simulation: Ansys Mechanical & Fluent, Finite Element Analysis, SolidWorks Simulation & Motion Analysis, hand-calculation tools (Excel)
Programming & Embedded systems: Python, MATLAB Simulink & Simscape, C, C++, Arduino
Other Tools: Excel, PowerPoint, PLM (Salesforce Propel), SolidWorks PDM
Standards & Testing: IEC 60068-2-6/-64/-31, IEC 60529 (IP68); test categories comparable to MIL-STD-810